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Low-Cost Versatile HAND MIXER

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Mixing liquids using a muddler or churner is a traditional technique. However, it is time consuming and sometimes fails to mix these properly. Here is a low-cost and versatile hand mixer circuit. It can be used for making stuff like *lassi, chhach* (buttermilk), butter and lemonade, among others. The author's prototype is shown in Fig. 1.

Fig. 1: Author's prototype

Circuit and working

The circuit diagram of the low-cost and versatile hand mixer is shown in Fig. 2. It is built around 5V voltage regulator 7805 (IC1), two BC547 transistors (T1 and T2), motor driver L293D (IC2), 12V geared motor (M1) and a few other components.

Fig. 3: A typical churner

Transistors T1 and T2 along with resistors R2 through R5 and capacitors C2 and C3 are configured as an astable multivibrator. Resistors R2 and R5 are connected as pull-up resistors. Resistors R3 and R4, and C2 and C3 are used for time period constants. You may change these values as per your requirement.

IC2 is used to drive motor M1. It is a dual H-bridge motor driver IC for driving the motor in clockwise as well as anti-clockwise directions.

IC1 (7805) is used as a power supply regulator. It gives constant 5V supply to power the astable multivibrator and IC2. Resistor R1 connected with LED1 is used to indicate power-on status. You can power up the circuit by connecting the power supply to connector CON1 using either 12V battery, 12V adaptor or 12V solar panel.

The churner is built around a DC geared motor. Its working is the same

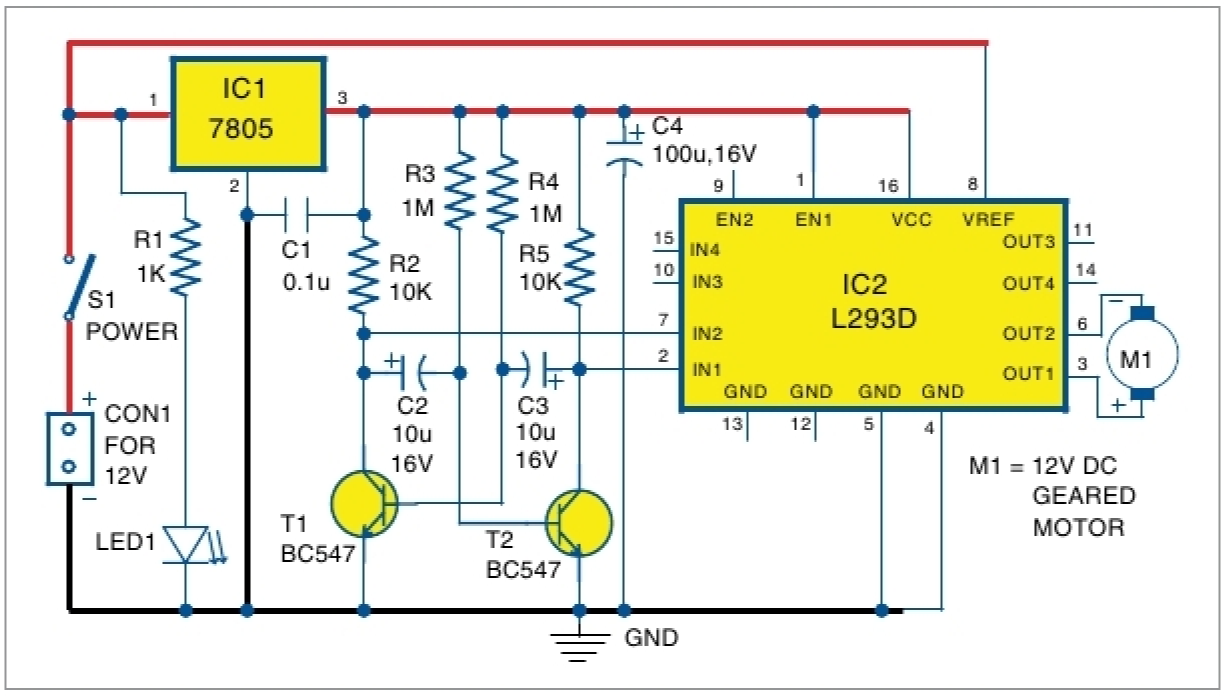

Fig. 2: Circuit diagram of low-cost and versatile hand mixer

PARTS LIST

Semiconductors:	
IC1	- 7805, 5V voltage regulator
IC2	- L293D motor driver
T1, T2	- BC547 npn transistor
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 1-kilo-ohm
R2, R5	- 10-kilo-ohm
R3, R4	- 1-mega-ohm
Capacitors:	
C1	- 0.1µF ceramic disk
C2, C3	- 10µF, 16V electrolytic
C4	- 100µF, 16V electrolytic
Miscellaneous:	
S1	- On/off switch
CON1	- 2-pin terminal connector
M1	- 12V, 500- to 1000-rpm geared motor
	- 12V battery/adaptor
	- Muddler/churner

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